

Superstore Sales Performance

A Visual Story of Growth, Profit Leaks, and Where to Act Next

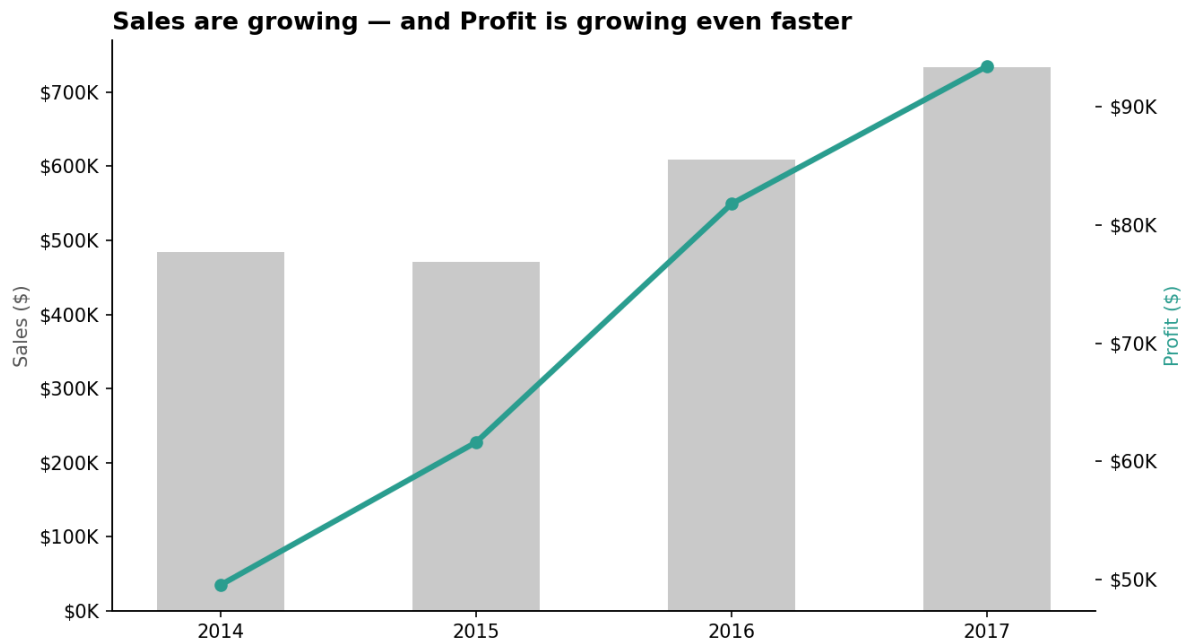
Data source: Sample Superstore dataset (9,994 orders, 2014-2017)

Prepared with: Python (Pandas, Matplotlib)

Executive Summary

Superstore's sales and profit have both grown steadily over four years, with profit growing faster than sales — a healthy sign. But that growth masks a problem: the **Furniture** category, and **Tables** in particular, are quietly losing money, driven by heavy discounting. Fixing discount policy on a small set of sub-categories and states could meaningfully lift overall profitability without touching revenue.

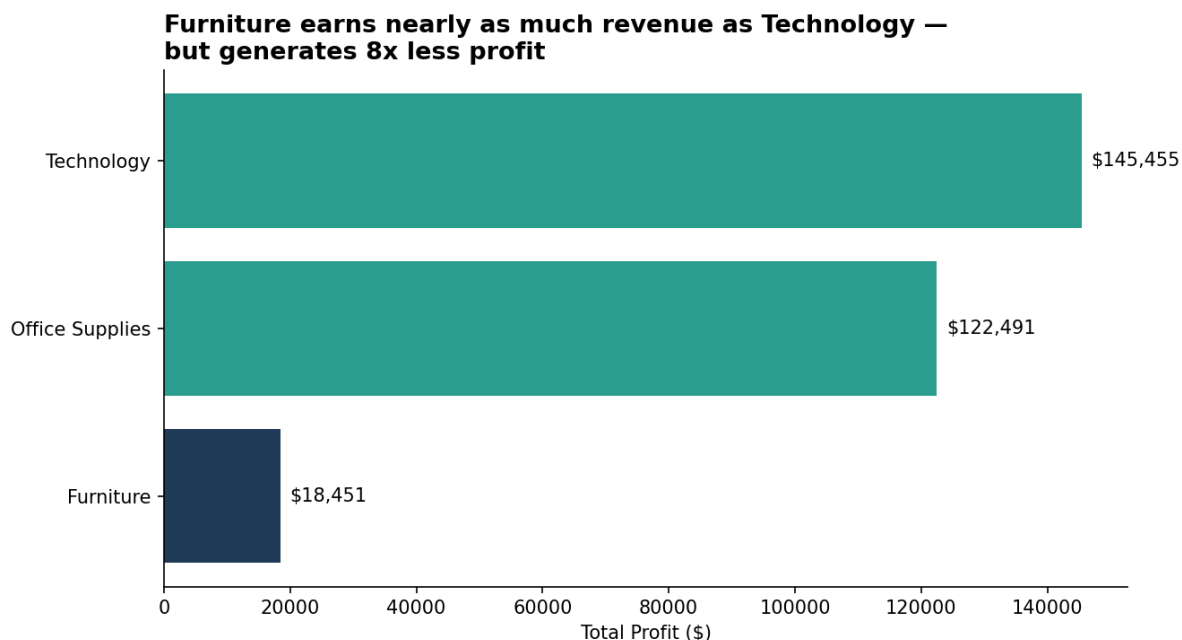
1. The Business Is Growing Year Over Year



Sales climbed from about \$484K in 2015 to \$733K in 2018, a 51% increase. Profit grew even faster over the same period, rising from roughly \$50K to \$93K — an 89% increase. This means the business isn't just selling more, it's converting that growth into profit more efficiently each year.

Key takeaway: Overall trajectory is healthy. The focus areas below are about tightening up specific leaks, not fixing a business in decline.

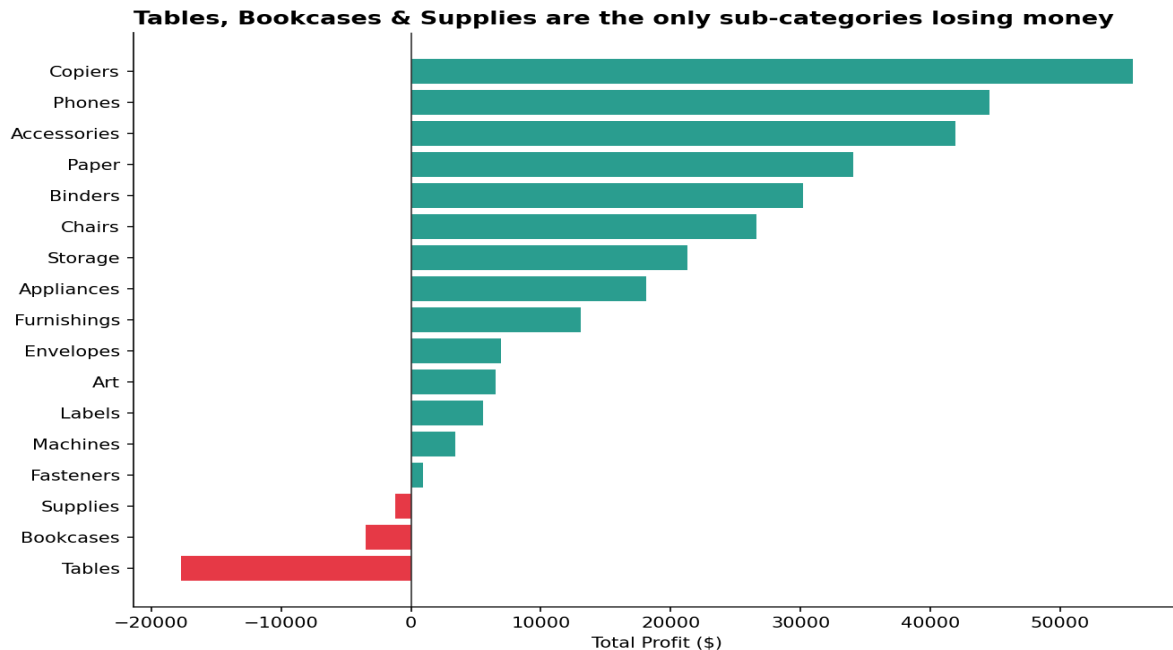
2. Furniture Is a Revenue Story, Not a Profit Story



Furniture generates \$742K in sales — almost as much as Technology's \$836K — yet it returns only \$18.5K in profit, compared to Technology's \$145K and Office Supplies' \$122K. Furniture is pulling its weight on the top line but not on the bottom line.

Key takeaway: Furniture needs a pricing/discount review, not a sales push. More Furniture sales at today's margins would barely move total profit.

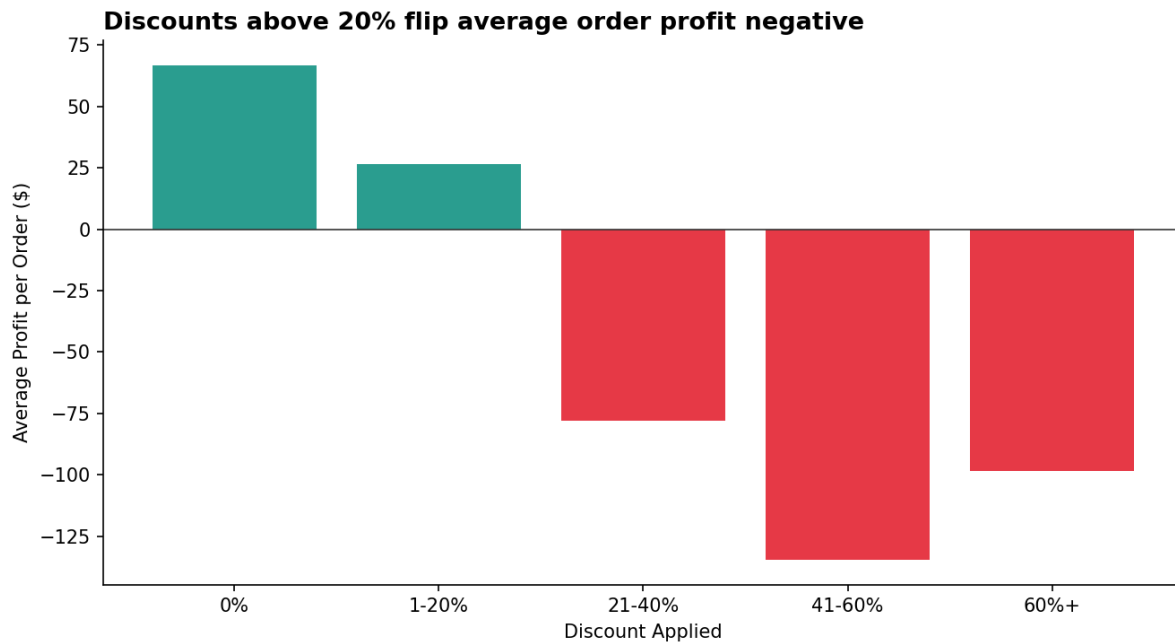
3. The Loss Is Concentrated in Just Two Sub-Categories



Drilling into sub-categories shows the Furniture problem is really a Tables problem: Tables alone lost \$17.7K, and Bookcases lost \$3.5K. Every other sub-category — including the rest of Furniture, like Chairs and Furnishings — is profitable. Supplies show a small, near break-even loss.

Key takeaway: This isn't a category-wide issue. Reviewing pricing and discount limits on Tables and Bookcases specifically would resolve most of Furniture's profit drag.

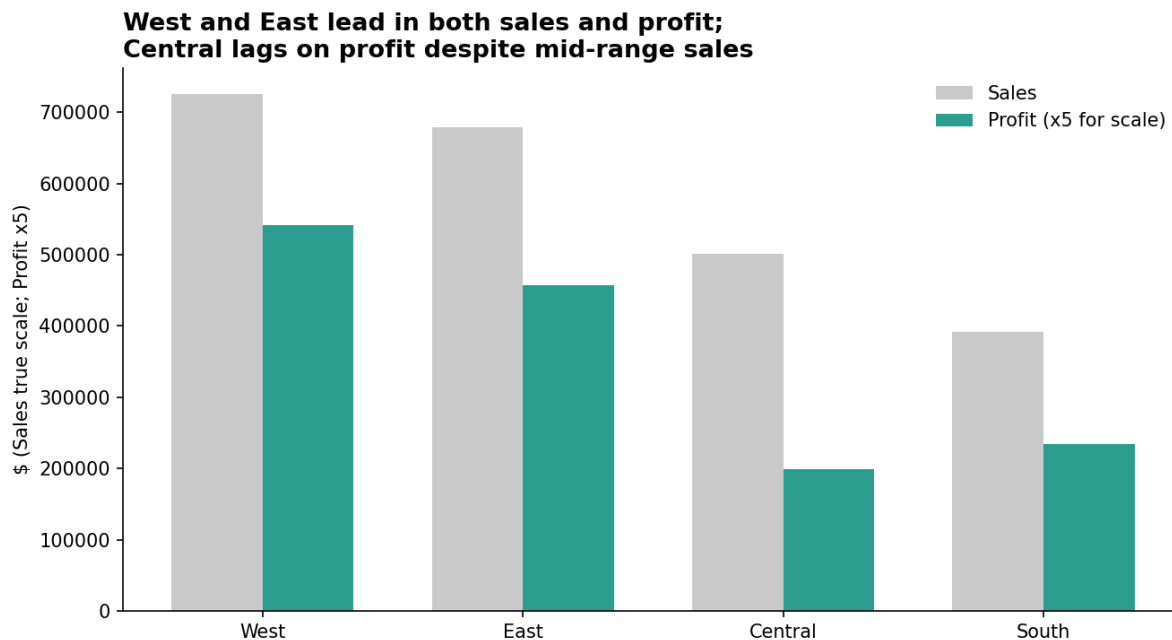
4. Discounting Above 20% Is Where Profit Turns Negative



Orders with no discount average \$67 profit each, and 1-20% discount still averages a positive \$27. Once discounts pass 20%, average profit per order goes negative — and it gets worse as discounts climb toward 60%+. Discounting correlates negatively with profit overall (correlation of -0.22).

Key takeaway: A 20% discount cap on standard orders — with exceptions requiring approval — would remove most of the discount-driven losses seen in Tables and Bookcases.

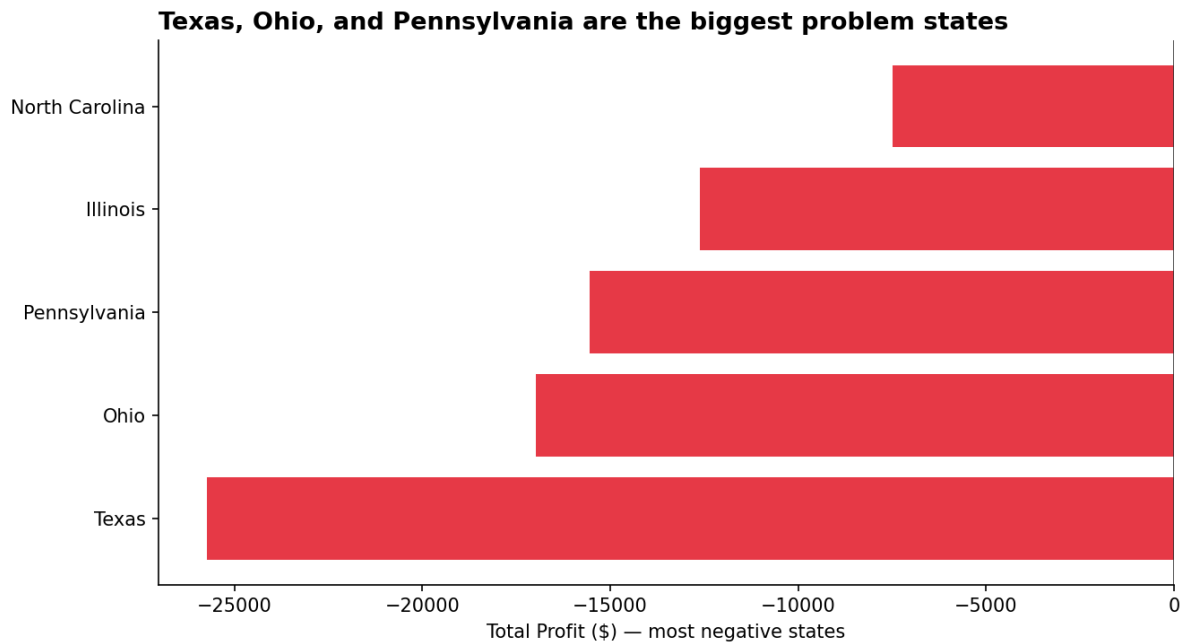
5. West and East Are the Strongest Regions



West and East lead on both sales and profit. Central shows mid-range sales but comparatively weaker profit, suggesting either heavier discounting or a less favorable product mix in that region.

Key takeaway: Central region's order-level discount and product mix is worth a closer look to close the gap with West and East.

6. A Handful of States Are Driving Furniture's Losses



Texas (-\$25.7K), Ohio (-\$17.0K), and Pennsylvania (-\$15.6K) are the three biggest loss-making states — together accounting for more loss than the Furniture category's total, meaning other states are actually offsetting some of this drag.

Key takeaway: Pairing the discount cap (insight 4) with tighter review of orders from Texas, Ohio, and Pennsylvania would directly target where the losses concentrate.

Summary Storyboard: The Business Case in 4 Beats

Beat	Insight	Recommended Action
1. Growth	Sales +51%, Profit +89% (2014-2017)	Keep current growth strategy — it's working
2. The Leak	Furniture drives revenue, not profit (\$18.5K profit vs. \$742K sales)	Review Furniture pricing strategy
3. The Root Cause	Tables & Bookcases lose money; losses climb sharply past 20% discount	Cap standard discounts at 20% for these sub-categories
4. Where It Hurts Most	Texas, Ohio & Pennsylvania are the top loss-making states	Prioritize discount policy rollout in these states first

Bottom line: Superstore doesn't have a sales problem — it has a discount discipline problem concentrated in two Furniture sub-categories and three states. Fixing this narrow issue protects existing revenue while directly improving company-wide profit margin.